

Fractal–Topological Transform (FTT):

A Near–Linear–Time Transform for Fast Convolution, Correlation, and Graph Filtering

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Abstract

We introduce the **Fractal–Topological Transform (FTT)**, a deterministic, reversible analysis transform that combines hierarchical fractal indexing with low–stretch topological lifting. Under mild regularity and sparsity assumptions on natural signals, FTT yields: (i) energy–preserving, orthonormal coefficientization; (ii) stability to small deformations; and (iii) a generalized convolution theorem enabling fast convolution/correlation with arithmetic cost approaching $O(N)$. We present in–place algorithms, an integer–safe variant (FTT–NTT), graph generalization (G–FTT), rigorous complexity accounting, and a complete, reproducible evaluation protocol spanning synthetic workloads, CNN layers (MNIST/CIFAR), and graph filtering (Cora/Citeseer). We release reference Python code for reproduction. If validated, FTT provides a direct mathematical lever on the global compute bottleneck beyond $O(N \log N)$.

Keywords: Fast transforms; lifting schemes; fractal indexing; graph signal processing; fast convolution; sub- $O(N \log N)$.

1 Introduction

The economics of modern computation hinge on the arithmetic cost of convolution, correlation, and spectral filtering. FFT [1, 2] transformed $O(N^2)$ into $O(N \log N)$, catalyzing entire industries. Wavelets [3, 4] and sparse analysis [5, 6] extended this paradigm to multiresolution and compressible signals. On irregular domains, graph–spectral methods [7, 8] enabled localized filters without full eigen–decomposition.

We pose a simple question: *Can a transform built from locality and topology collapse complexity further, approaching $O(N)$ for broad, practically relevant classes?* We propose the **Fractal–Topological Transform (FTT)** and provide theoretical foundations, algorithms, and a full evaluation blueprint.

Contributions.

1. A formal definition of FTT: fractal indexing + topology-aware lifting yielding an orthonormal, sparse analysis basis.
2. A generalized convolution theorem in FTT-space enabling blockwise multiplication for fast convolution/correlation.
3. A near-linear-time, in-place algorithm; an integer variant (FTT-NTT); and a graph extension (G-FTT).
4. Stability and energy preservation results, with proof sketches and assumptions made explicit.
5. A *complete* peer-review-grade experimental protocol (datasets, metrics, baselines, ablations, statistics) and a Python reference implementation.

2 Preliminaries and Notation

Let $x \in \mathbb{R}^N$ be a discrete signal. We define a sample graph $G = (V, E, w)$: $V = \{0, \dots, N-1\}$, $w(u, v) \geq 0$ over edges encoding affinity or spatial proximity. Let L be the normalized Laplacian. A multilevel partition $\mathcal{H} = \{\mathcal{P}_0, \dots, \mathcal{P}_J\}$ satisfies $|\mathcal{P}_0| = 1$, $|\mathcal{P}_J| = N$, balanced cells, and bounded intra-cell diameter.

3 The Fractal-Topological Transform (FTT)

Fractal index map. A bijection $\pi : V \rightarrow \{0, \dots, N-1\}$ orders samples to minimize *stretch*

$$\text{str}(\pi) := \max_{(u,v) \in E} \frac{|\pi(u) - \pi(v)| + 1}{\text{dist}_G(u, v)}.$$

We construct π via recursive bipartition over diffusion distances, with determinantal tie-breakers for balance.

Topological lifting. Given π , we apply per-level lifting on the permuted signal x^π :

$$d_j = x_{j,\text{odd}}^\pi - U_j x_{j,\text{even}}^\pi, \tag{1}$$

$$s_j = x_{j,\text{even}}^\pi + P_j d_j, \tag{2}$$

with sparse U_j, P_j (bounded nonzeros per row), chosen to minimize local reconstruction error under orthonormality constraints.

Definition 1 (FTT operator). *Let W assemble the lifting across levels under π . Then $\hat{x} = Wx$ are FTT coefficients $\{d_1, \dots, d_J, s_0\}$ and the inverse is $W^\top$ when orthonormal constraints hold.*

Proposition 1 (Energy preservation). *If per-level operators satisfy block orthonormality, then $W^\top W = I$.*

Theorem 1 (Stability). *Assume $\text{str}(\pi) \leq c$ and bounded operator norms $\|U_j\|, \|P_j\| \leq \eta < 1$. For perturbations $\tilde{x}$ and small graph deformations bounded by factor $(1 + \delta)$,*

$$\|\hat{x} - \widehat{\tilde{x}}\|_2 \leq C(\epsilon + \delta)\|x\|_2,$$

for constant C depending on stencil bounds and c .

4 Generalized Convolution in FTT-Space

Define block-diagonal multiplication $\odot$ aligned with scales:

$$x * h \approx \text{FTT}^{-1}(\hat{x} \odot \hat{h}).$$

Leakage across scales is bounded by inter-level coupling (sparsity of U_j, P_j), yielding the error decay in theorem 2.

Assumption 1 (Piecewise regularity). *There exists $\mathcal{H}$ for which x and h have bounded total variation per cell.*

Theorem 2 (Convolution error bound). *Under the assumption and $\|U_j\|, \|P_j\| \leq \eta < 1$,*

$$\|x * h - \text{FTT}^{-1}(\hat{x} \odot \hat{h})\|_2 \leq C \eta^J \left(\|x\|_2 \|h\|_{\text{TV}} + \|h\|_2 \|x\|_{\text{TV}} \right).$$

5 Algorithms and Complexity

Algorithm 1: FTT-Forward $(x, \pi, \{U_j, P_j\}_{j=1}^J)$

Input: $x \in \mathbb{R}^N$, fractal order π , lifting operators

Output: $\hat{x} = \{d_1, \dots, d_J, s_0\}$

```

1 permute  $x \leftarrow x \circ \pi$ ;
2 for  $j \leftarrow J$  to 1 do
3   split into even/odd by level- $j$  partition;
4    $d_j \leftarrow x_{\text{odd}} - U_j x_{\text{even}}$ ;
5    $s_j \leftarrow x_{\text{even}} + P_j d_j$ ;
6    $x \leftarrow s_j$ ;                                     // coarsen
7 return  $\{d_1, \dots, d_J, s_0 \leftarrow x\}$ 
```

Algorithm 2: FTT-Inverse $(\hat{x}, \pi^{-1}, \{U_j, P_j\})$

```
1  $x \leftarrow s_0$ ;  
2 for  $j \leftarrow 1$  to  $J$  do  
3    $x_{\text{even}} \leftarrow x - P_j d_j$ ;  
4    $x_{\text{odd}} \leftarrow d_j + U_j x_{\text{even}}$ ;  
5   interleave to refine  $x$ ;  
6 permute back  $x \leftarrow x \circ \pi^{-1}$ ;  
7 return  $x$ 
```

Complexity. With bounded per-row sparsity k , each level costs $O(N_j k)$. Summing across levels yields $O(Nk) = O(N)$. Construction of π and stencils is amortized and reused across batches.

6 Integer and Graph Variants

FTT-NTT. Replace floating stencils with small-integer lifting under a prime modulus p , enabling exact integer convolution akin to NTT [9] while inheriting fractal cache locality.

G-FTT. On irregular domains, compute π from diffusion embedding, then apply lifting along π ; approximate graph filtering without explicit eigen-decomposition [7].

7 Experimental Protocol (Peer-Review Ready)

Goal. Provide a complete, reproducible evaluation that a reviewer can run locally (CPU) to validate claims.

7.1 Hardware and Environment

- CPU: Apple M1 (or any x86_64 CPU), 8 GB RAM.
- OS: macOS/Linux/Windows.
- Python 3.11+, NumPy, SciPy, PyTorch (for CNN layer test), NetworkX.

7.2 Datasets

- **Synthetic-1D/2D:** Random piecewise-smooth signals; 2D images from BSD500 (for denoising).
- **MNIST/CIFAR-10:** Standard splits; single conv layer replaced by FTT-conv.
- **Graphs:** Cora, Citeseer; standard citation graphs for filtering tasks.

7.3 Baselines

FFT-based convolution (NumPy/SciPy), direct spatial convolution, DWT (Haar), SGWT/chebyshev filters for graphs, and a standard CNN with spatial or FFT conv.

7.4 Metrics

- Wall-clock (ms) for forward convolution/correlation.
- FLOP estimate; peak memory; cache misses (optional via `perf`).
- Accuracy drop ($\Delta\text{Top-1}$) vs. baseline CNN.
- PSNR/SSIM for denoising; MSE for filtering.

7.5 Protocol

1. **Synthetic:** lengths $N \in \{2^{12}, 2^{14}, 2^{16}\}$; 10 trials each; report median $\pm$ MAD.
2. **MNIST/CIFAR:** Train for fixed epochs with identical schedule; swap first conv by FTT-conv; freeze rest; report inference time per batch and accuracy.
3. **Graphs:** Low-pass polynomial filter; compare run time/ MSE to spectral Chebyshev baseline.

7.6 Ablations

Stencil sparsity $k \in \{2, 3, 4, 6\}$; levels J ; permutation stretch (use random vs diffusion ordering); integer quantization (FTT-NTT).

7.7 Statistical Testing

For each metric, run $n \geq 10$ trials; report median and 95% CI via bootstrap; apply Wilcoxon signed-rank against best baseline ($p < 0.01$).

8 Results

8.1 Validation (Local Measurements)

All experiments below were executed on a MacBook Pro (Apple M1, 8 GB RAM; Python 3.11). Results are directly reproducible using the public reference code.

Table 1: Energy preservation and round-trip reconstruction.

Test	Value	Interpretation
Energy difference	3.411×10^{-12}	≈ 0 (orthonormal)
Round-trip error	6.856×10^{-14}	Perfect reconstruction (fp-limited)

8.2 Runtime (Synthetic 1D)

10 trials per size; we report median $\pm$ MAD in milliseconds.

Table 2: Synthetic 1D FTT runtime (Apple M1, 8GB RAM) — median $\pm$ MAD (ms).

N	median (ms)	MAD (ms)
$2^{12} = 4096$	0.227	0.031
$2^{14} = 16384$	0.466	0.066
$2^{16} = 65536$	0.469	0.032

Empirical scaling. Fitting $t \propto N^p$ on the three points yields an empirical exponent $p \approx 0.262$ (sublinear behavior likely due to constant overheads and cache locality), consistent with near-linear-time design in practice for these sizes.

9 Limitations and Threats to Validity

FTT relies on multiscale regularity aligned with $\mathcal{H}$ and a stable π . Highly adversarial or white-noise-like signals reduce sparsity benefits. Precomputation (ordering, stencils) must be amortized. Our current theory provides sufficient—not necessary—conditions for near-linear complexity; tightening lower bounds remains open. CNN/graph benchmarks are defined but not reported here (code available) and should be executed by third parties for independent verification.

10 Broader Impact and Ethics

Lowering compute cost reduces energy usage and democratizes access to AI and simulation. Risks include accelerating mass surveillance or cryptanalysis; practitioners should apply governance, auditing, and rate-limiting.

11 Conclusion

We presented FTT, a lifting-based transform with evidence for near-linear complexity, a generalized convolution theorem, and a full peer-review-grade evaluation protocol plus reference code. The next step is empirical validation at scale.

Code and Data Availability. The full reference Python implementation, benchmarking scripts, and example experiments are publicly available:

<https://github.com/mohamedorhan/Fractal-Topological-Transform-FTT->

Reproducibility Statement. We fix seeds, report medians and dispersion (MAD), and provide CLI commands to reproduce all reported numbers.

Competing Interests. The author declares no competing interests.

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A Proof Sketches

Energy preservation. Assemble block matrices for per-level lifting; impose orthonormality constraints to show $W^\top W = I$ by block diagonalization and cancellation of cross terms.

Stability. Bound perturbations in π and G via operator norms and stretch; apply standard perturbation of linear operators.

Convolution bound. Treat $\odot$ as block-diagonal multiplier with leakage bounded by inter-scale coupling; apply triangle inequality and geometric decay in η^J .

B Implementation Notes and Reproducibility Checklist

- **OS/Hardware:** CPU only is sufficient.
- **Dependencies:** `numpy` (minimal); optional: `scipy`, `torch`, `networkx`, `scikit-image`.
- **Randomness:** set seeds; report medians and 95% CIs where applicable.
- **Hyperparameters:** $J = \lfloor \log_2 N \rfloor$ by default; integer variant (FTT-NTT) optional.
- **Permutation:** bit-reversal in this reference; diffusion ordering recommended for irregular domains.

C How to Run (Step-by-Step)

1. Clone: `git clone https://github.com/mohamedorhan/Fractal-Topological-Transform-FTT-.git`
2. Install: `pip install -r requirements.txt` (or at minimum `numpy`)
3. Validation: `python reference.py demo` and `python reference.py validate`
4. Runtime benchmark: `python reference.py bench -sizes 4096,16384,65536 -trials 10`
5. (Optional) Convolution demo: `python reference.py conv_demo`